

ENVIRONMENTAL HACKS 2026 · BHARAT BUILDS TOUR · TRACK 01: AIR

HawaCommand

An AI command centre that finds, forecasts and fixes Delhi's pollution hotspots, and proves every fix worked.

Full technical and execution proposal · hardware specification · software architecture · impact and business case

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Built on AWS · Strands Agents SDK

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THE IDEA ON ONE PAGE

1 Executive summary

> **Delhi does not lack anti-pollution machines. It lacks a brain to direct them and a way to prove they work.** HawaCommand is that brain: it senses the air street by street, finds where pollution is coming from, forecasts where it will hurt the most people, sends the city's existing sprinklers, smog guns and inspectors there before the spike, warns the schools in the path, and then measures whether each action actually cleaned the air.

The problem in one line

Delhi measures its air too sparsely, reacts after the damage, deploys a large anti-pollution fleet on fixed routes, never checks whether those actions worked, and ranks places by AQI instead of by how many people are exposed.

The solution in one line

A closed loop of **Sense → Map → Attribute → Forecast → Decide → Act → Verify**, running every three hours on AWS, fed by low-cost sensors that ride on vehicles already moving through the city and by a tracker on every cleaning machine.

What makes it different

Most teams will build	HawaCommand builds
An AQI dashboard or alert app that **shows the number**	A command system that **acts on the number and proves the result**
Ranking by AQI	Ranking by **people exposed × vulnerability** (schools, hospitals, outdoor workers)

Most teams will build	HawaCommand builds
"Air is bad"	**Why** it is bad (dust, traffic, burning, regional smoke) and which tool fixes that cause
Reacting to today	**72-hour forecasts** and action 24–48 hours before the spike, as China's early-action studies recommend
No proof	**Causal before/after verification** of every sprinkler run, using a tracker on the machine itself

Borrowed from Asia, rebuilt for Delhi

[CHINA] Hotspot-grid management, taxi-mounted mobile sensing, 72-hour street-scale forecasting, graded emergency response and performance-graded polluters. **[JAPAN]** Open real-time public data (AEROS), pollutant-specific targeted warnings, and Tokyo's lesson that measured, enforceable rules on the biggest source beat symbolic hardware. **[ORIGINAL]** Intervention sensing, causal verification, anti-gaming checks and exposure-first multi-agent planning.

Why it can win

- **Idea and impact:** targets a documented gap in Delhi's own programme (unverified dust-control spending).
- **Built on AWS:** IoT Core, Lambda, Timestream, RDS/PostGIS, SageMaker, Bedrock, plus the open-source Strands Agents SDK.
- **Usability:** officers approve plans in one click; drivers get a route; schools get a WhatsApp message in Hindi or English.
- **Execution:** the whole cloud pipeline runs for real; the device fleet is simulated in the exact firmware packet format.
- **Cost:** a 100-bus district pilot costs a small fraction of one smog tower (Section 12).

WHAT IS ACTUALLY BROKEN

2 The problem

Picture a November morning in Delhi. The phone says **AQI 380**. That single number hides five separate failures.

2.1 The city is half-blind

Delhi had **47 continuous monitoring stations** in May 2026 for about **1,484 km²**, roughly one station per 31 km². The government's own target is one per 25 km². A station represents the air within a few hundred metres to a few kilometres, but pollution changes street to street: a construction site or a burning garbage heap can push one lane far above the reading of a station 3 km away. A CAG audit also found stations sited near trees, major roads and unpaved roads, which makes some readings unrepresentative.

2.2 Everyone reacts after the damage

Alerts arrive when the air is already bad: children are already at the bus stop and workers are already on the road. Research on Beijing's 2015 red alerts found that starting emission cuts **one to two days before** an episode lowered the peak significantly, compared with starting on the day.

2.3 A large fleet works on guesswork

Delhi has mobilised **376 anti-smog guns, 266 water sprinklers and 91 mechanised road sweepers**, all GPS-fitted, plus around 2,000 enforcement teams. GPS proves where a machine went. It does not prove the air got cleaner. Routes are largely fixed, while hotspots move with wind, traffic and construction.

2.4 Nobody checks whether it worked

CEEW notes that in winter, road dust is about **15% of ambient PM2.5**, yet about **64% of available NCAP funds** go to road-dust mitigation, and there is limited evidence on how well anti-smog guns work. An IIT Delhi finding reported that water spraying had a very small effect on PM2.5, limited to about 200 m. Money is spent without a feedback signal.

2.5 "Worst AQI" is not "most people hurt"

AQI 450 over an empty industrial yard harms fewer people than AQI 350 next to three schools and a hospital at 8 AM. Nobody ranks hotspots by human exposure.

> **Problem statement.** Delhi measures its air too sparsely, acts too late, deploys its anti-pollution fleet without a targeting brain, never verifies whether an action worked, and does not prioritise the people most exposed. The result: large spending, weak evidence, and children breathing the worst air at the worst hour.

Who suffers

Group	What they face today
School children	Morning assembly and sports at peak-pollution hours; no street-level warning for their school
Outdoor workers, vendors, riders	Hours of exposure on the dirtiest roads; no route or timing advice
City officers (MCD, DPCC)	Must deploy hundreds of machines daily with station averages and no effectiveness data
Taxpayers	Crores spent on dust control and smog towers with little measured return

THE ORIGINAL IDEA, AND WHAT WE KEPT

3 Why not flying purifier bots

The first version of this idea was a fleet of about 2,000 flying AI capsules that would sense the air, send data to the cloud, and release chemicals to purify hotspots. The instinct (**hyperlocal sensing → cloud AI → targeted action**) is exactly right. The flying purifier is not.

Issue	Reality check
Air volume	Delhi's winter mixing layer holds on the order of hundreds of billions of m^3 of air, replaced by wind within hours. Even thousands of purifiers move a tiny fraction of it.
Evidence from Delhi	The DPCC told the National Green Tribunal that smog towers are ineffective, that more than 40,000 would be needed, and that each costs about ₹15 lakh a month to run. A small drone filter is far weaker than a 24 m tower.
Chemicals in open air	No approved spray safely neutralises PM2.5 over people; a spray is a new aerosol that residents breathe.
Airspace and batteries	Much of central Delhi is restricted airspace; flight needs DGCA permissions; batteries last minutes, not hours.
Hackathon judging	Unbuildable in the event window, no clear user, and "big problem solved vaguely" is explicitly penalised.

What we kept, and what replaced the drones

Original idea	HawaCommand
2,000 custom drones sensing the air	Low-cost sensors riding on buses, e-rickshaws and delivery bikes that already move everywhere
Cloud AI decides where to act	Same, made concrete: digital twin, source fingerprinting, 72 h forecast, exposure-first planner
Signal sent to bots to purify	Signal sent to the city's real sprinklers, smog guns, sweepers and inspection teams

Original idea	HawaCommand
Chemical release	Dust suppression where dust is the cause, source enforcement where burning is the cause, protection where smoke is regional
No proof it works	Tracker on every machine plus causal before/after verification

PROVEN IDEAS WE BORROW, ADAPTED FOR DELHI

4 What China and Japan already learned

China and Japan fought severe urban air pollution before India did. We studied what they built, what worked, and what backfired. Each lesson below maps to a HawaCommand feature in Section 6.

4.1 China

System	What they did	Result / lesson	Our feature
[CHINA] **Hotspot grid management**	Covered 28 cities in the Beijing–Tianjin–Hebei region with 3×3 km blocks (36,793 total) and used satellite and weather analysis to pick about 3,600 high-PM2.5 blocks for close monitoring. Beijing added ~1,500 micro-stations and mobile monitors.	In the Cangzhou pilot, PM2.5 in the pilot area fell 38.3%, versus about 20% city average; 6,325 polluting enterprises identified.	F1 Hotspot Grid Engine
[CHINA] **Taxi-mounted sensor networks**	Shanghai put CO, NO_2 and PM2.5 sensors on 125 taxis; Guangzhou on 314 taxis; Jinan combined 200 mobile vehicles with 614 fixed micro-stations.	80–100 taxis covered ~80% of major roads at ~200 m resolution; Jinan reconstructed 500 m, hourly PM2.5 maps. Guangzhou calibrated against roadside reference stations.	F2 Mobile sensing, F3 Drive-by calibration, F4 Digital twin
[CHINA] **IBM Green Horizons, Beijing**	Forecasting and decision-support system with 1 km × 1 km forecasts 72 hours ahead, plus 10-day trends, combining models with adaptive machine learning.	Reported ~30% better precision than conventional approaches. A Delhi agreement targeted traffic–pollution links and "what-if" scenario modelling.	F5 72 h forecaster, F13 What-If simulator
[CHINA] **Graded heavy-pollution response**	Blue/yellow/orange/red alerts issued at least 24 h ahead from forecasts; each level triggers preset emission cuts. Industries and construction sites are performance-graded, and cleaner (grade A) firms face lighter restrictions.	Avoids one-size-fits-all shutdowns. Enforcement uses source monitoring, industrial electricity use, traffic flow, satellite data and the hotspot grid.	F6 Graded Response Copilot, F9 Site Compliance Score, F10 Power-proxy check
[CHINA] **Smoky-vehicle AI**	Deep-learning detection of visibly smoky vehicles from roadside CCTV; roadside remote-sensing of exhaust used in 70+ cities by 2016.	Far cheaper than optical remote-sensing rigs; the hard part is telling smoke from shadows and wet roads.	F8 Smoky Vehicle Spotter
[CHINA] **Warning: monitor gaming**	Research on China's monitoring expansion found improvement concentrated in monitored areas without a matching city-wide drop.	Local cleaning near monitors can make readings look better than the city really is.	F15 Anti-gaming guard

4.2 Japan

System	What they did	Result / lesson	Our feature
[JAPAN] **AEROS "Soramame-kun"**	National system publishing real-time PM2.5, SPM, oxidants, NO_2, SO_2, CO and more, openly, from every monitoring station.	Open data built public trust and powers apps and research.	F21 Open Air Data portal

System	What they did	Result / lesson	Our feature
[JAPAN] **Pollutant-specific warnings**	Japan uses per-pollutant standards and targeted alerts (for example the photochemical-oxidant warning) instead of one composite index.	People get advice that matches the actual threat.	F16 Targeted alerts
[JAPAN] **Tokyo "Operation No Diesel" (2003)**	Older diesel trucks and buses had to fit certified particulate filters or catalysts, or be kept out of the city.	Roadside SPM fell by over 20% between 2002 and 2004; stations meeting the national standard rose from 0% to 97%.	Lesson for F7 + F22: attribute sources, then act on the biggest measured one

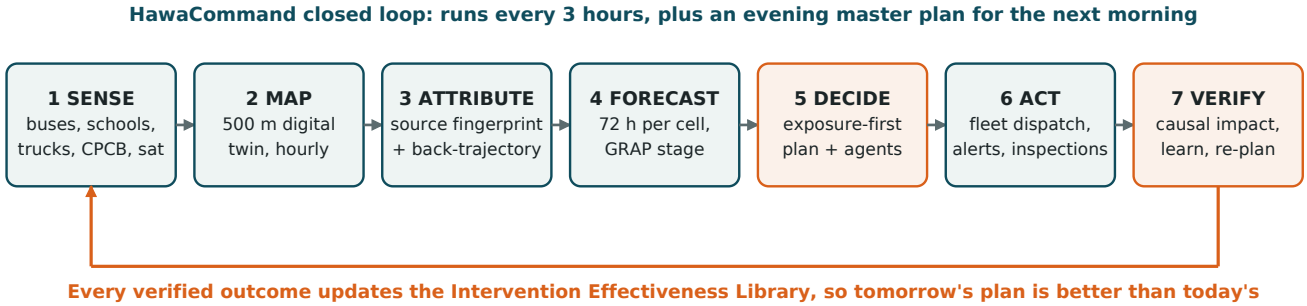
4.3 What we refuse to copy

Smog towers. Delhi's towers were inspired by the one in Xi'an, China. IIT-Bombay measurements summarised by DPCC showed effect falling to about 12–16% beyond 100–200 m, and the Connaught Place tower cost about ₹20 crore to build. HawaCommand copies China's *data and governance systems*, not its purifier hardware.

THE CLOSED LOOP, IN SIMPLE WORDS

5 The solution: HawaCommand

One line. HawaCommand sees Delhi's air street by street, works out where each hotspot's pollution is coming from, predicts where it will hurt the most people in the next 72 hours, tells the city exactly where to send its sprinklers, smog guns and inspectors, warns the schools in the path, and then proves which actions actually cleaned the air.



5.1 The seven steps, explained simply

Step	What happens	Simple analogy
1 Sense	Sensors on buses and bikes, school sentinels, trackers on cleaning trucks, official stations, weather and satellite fire data all stream to AWS.	Official stations are CCTV at a few crossings; our buses are police walking every street.
2 Map	The city is cut into 500 m cells. A digital twin fills gaps between sensors using statistics plus wind drift, and gives every cell a value and a confidence.	Weather radar for pollution.
3 Attribute	Each hotspot gets a likely cause (road dust, traffic, burning, regional stubble smoke) and an upwind suspect location.	A doctor diagnosing before prescribing.
4 Forecast	72-hour forecast per 1 km cell, and the GRAP stage it will trigger.	A cyclone warning, but for smog.
5 Decide	Planner agents rank cells by people exposed and assign machines to causes they can fix; an Auditor agent challenges the plan; an officer approves.	A war room with a very fast analyst.

Step	What happens	Simple analogy
6 Act	Drivers receive routes; inspectors receive targets; schools and workers receive warnings.	Dispatch in an ambulance service.
7 Verify	The tracker logs exactly when a truck sprayed; the system compares the cell against similar untreated cells and records the true effect.	A clinical trial for every sprinkler run.

5.2 Goals

- **See:** street-level coverage of a pilot district, hourly, with a confidence score for every cell.
- **Warn early:** forecast spikes 24–72 hours ahead and trigger action before GRAP stages kick in.
- **Target:** reduce more person-hours of exposure with the same fleet (the What-If simulator measures this).
- **Prove:** a measured effect, with uncertainty, for every intervention, and a library of what works where.
- **Protect:** vulnerable places warned before the spike, with a concrete safe window.

22 FEATURES, TAGGED BY ORIGIN AND BUILD TIER

6 Complete feature list

Tiers: **BUILD** = working in the hackathon demo · **STRETCH** = build if time allows · **VISION** = pitched with architecture only.
Origin: **[CHINA]** **[JAPAN]** **[ORIGINAL]**.

#	Feature	Origin	Tier	What it does
	SENSE AND MAP			
F1	**Hotspot Grid Engine**	[CHINA]	BUILD	Delhi split into 1 km (planning) and 500 m (map) cells. Satellite AOD, fire points and station history pre-rank "key cells" for dense monitoring, like China's 3×3 km grid but finer.
F2	**Mobile sensing fleet**	[CHINA]	BUILD	HawaPod-M sensors on DTC and cluster buses, e-rickshaws and delivery bikes. One bus covering 50 km a day produces thousands of geo-tagged readings.
F3	**Drive-by self-calibration**	[CHINA]	BUILD	Every time a bus passes within 200 m of a reference station in stable conditions, the device's humidity-aware correction curve is re-fitted automatically. No manual calibration visits.
F4	**Air digital twin**	[CHINA]	BUILD	Gaussian-process interpolation plus wind-driven advection produces a 500 m, hourly map with a confidence value per cell (Jinan-style fixed + mobile fusion).
F5	**72 h forecaster**	[CHINA]	BUILD	Gradient-boosted model per cell using lagged readings, IMD/ERA5 weather, calendar and fire features; outputs P10/P50/P90 so officers see uncertainty.
F11	**Active sensing router**	[ORIGINAL]	STRETCH	Low-confidence cells are sent to the planner as "blind spots"; the system suggests which bus route or rider should pass through next.
	UNDERSTAND AND DECIDE			
F6	**Graded Response Copilot**	[CHINA]	BUILD	Maps forecasts to Delhi's GRAP stages 24–48 h ahead and generates a ward-wise checklist of stage actions, so agencies act before the stage is formally invoked.
F7	**Source fingerprint + back-trajectory**	[ORIGINAL]	BUILD	PM2.5/PM10 ratio, NO_2/CO co-signals, VOC, time and fire points classify each hotspot's cause; reversing wind vectors pins a likely upwind source.
F12	**Exposure engine**	[ORIGINAL]	BUILD	Person-hours of harmful exposure per cell = concentration × population × time-of-day weights × vulnerability (schools, hospitals, markets, bus stops).
F13	**Dispatch optimiser + What-If**	[CHINA]	BUILD	OR-Tools vehicle routing assigns each machine to causes it can fix. The What-If screen compares "usual routes" against the AI plan in person-hours avoided.

#	Feature	Origin	Tier	What it does
F14	**Multi-agent command room**	[ORIGINAL]	BUILD	Strands agents: Forecaster, Planner, Auditor (argues against the plan), Communicator. Nothing dispatches without officer approval.
F9	**Site Compliance Score**	[CHINA]	STRETCH	Construction sites and units graded A/B/C from sensor-measured behaviour; better grades face lighter curbs on bad days, rewarding good actors instead of blanket bans.
F10	**Power-proxy compliance**	[CHINA]	VISION	On restriction days, smart-meter load of listed units checks whether a promised shutdown really happened.
	ACT AND VERIFY			
F17	**HawaTag intervention tracking**	[ORIGINAL]	BUILD	Flow sensor and pump-current sensor on each sprinkler or smog gun log exactly when and where it sprayed, with PM readings before and after.
F18	**Causal verification + effectiveness library**	[ORIGINAL]	BUILD	Synthetic-control comparison per intervention gives effect size, duration and confidence; results feed back into the planner.
F15	**Anti-gaming guard**	[CHINA]	STRETCH	Flags cells where cleaning clusters around official stations while mobile readings nearby stay high, China's documented failure mode.
F8	**Smoky Vehicle Spotter**	[CHINA]	STRETCH	Optional HawaEye camera node runs a YOLO-class model to flag visibly smoky vehicles; evidence goes to an authorised officer queue for human review.
	PROTECT AND OPEN UP			
F16	**Targeted alerts**	[JAPAN]	BUILD	Alerts name the dominant pollutant and give matching advice (masks and indoors for PM2.5 smoke; avoid the road for traffic NO_2), in Hindi and English, via WhatsApp and SMS.
F19	**School safe-window scheduler**	[ORIGINAL]	STRETCH	Indoor/outdoor HawaPod-S pairs plus the forecast give each school the safest outdoor window and tell it when to seal or ventilate classrooms.
F20	**Citizen Eyes**	[ORIGINAL]	STRETCH	WhatsApp photo of burning → multimodal model checks it → matched to a sensor spike → inspection dispatched → reporter told the outcome.
F21	**Open Air Data portal**	[JAPAN]	STRETCH	AEROS-style public map and API of calibrated street-level data, with methods published.
F22	**Evidence report**	[ORIGINAL]	VISION	Weekly auto-generated PDF: hotspots, sources, actions, measured impact per rupee, and recommended reallocation.

Feature numbers are stable IDs used in the rest of this document and in the codebase; they are grouped by stage, not in numeric order.

WHO TOUCHES THE SYSTEM AND HOW

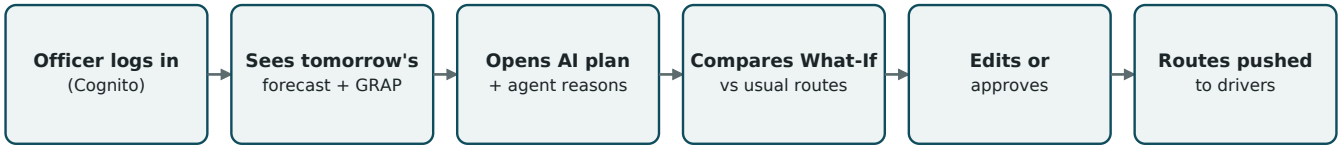
7 Users and user flows

User	Needs	Sees	Does
Ward / district officer (MCD, DPCC)	Where to send machines and teams, and proof it worked	Live twin map, 72 h forecast, GRAP outlook, AI plan with reasons, What-If comparison, impact report	Approves or edits the plan; escalates inspections
Driver / field team	A clear route and nothing else	Ordered stops on a phone PWA or WhatsApp, with arrival windows	Drives the route; HawaTag confirms the spray automatically
Inspector	A target, not an area	Suspected source pin, evidence (sensor spike, citizen photo, trajectory)	Visits, records outcome
School principal	When is it safe outdoors	WhatsApp alert with safe window; indoor vs outdoor readings	Moves assembly or sport; seals or ventilates classrooms
Citizen / outdoor worker	Should I avoid this road now	Pollutant-specific alert, cleaner-route tip	Reports burning with a photo

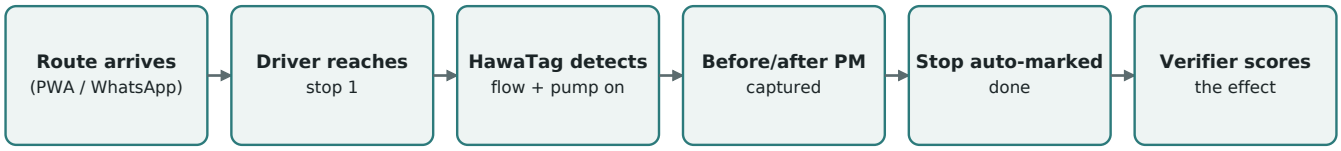
User	Needs	Sees	Does
Site manager (construction)	Stay compliant, avoid shutdowns	Own Site Compliance Score and PM10 trend	Fixes dust control; earns a better grade

7.1 Key flows

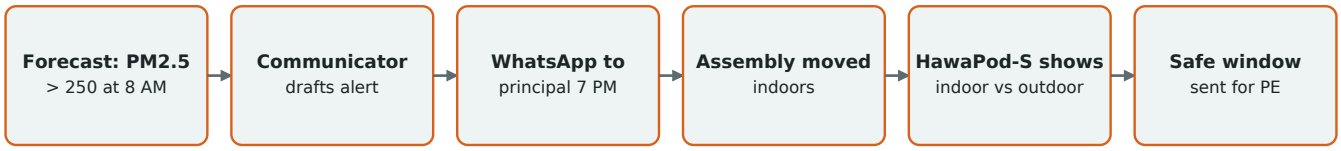
Flow A · Officer approves tomorrow's plan



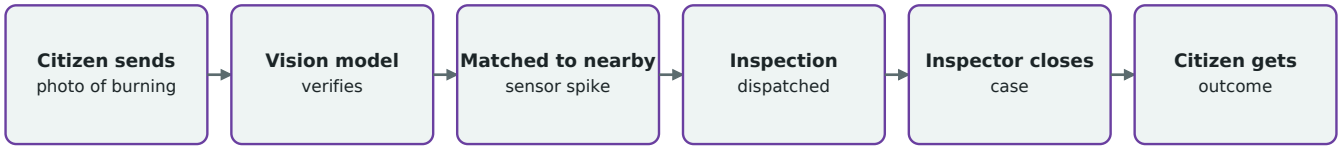
Flow B · Driver runs a route (no manual reporting)



Flow C · School protection



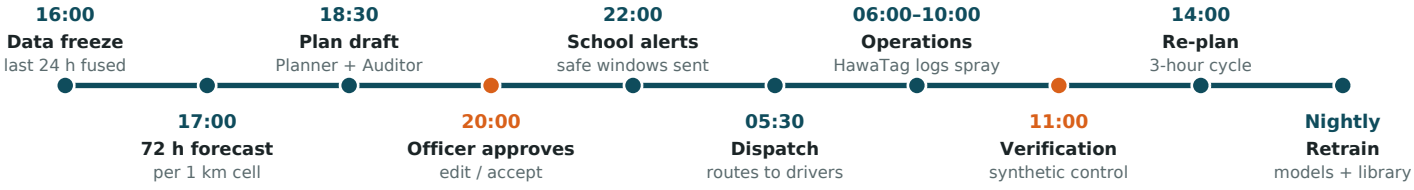
Flow D · Citizen Eyes



HOW THE SYSTEM RUNS, HOUR BY HOUR

8 Execution flow

A day in HawaCommand (winter operating mode)



8.1 Scheduled cycle (EventBridge + Step Functions)

Stage	Trigger	Work done	Output
Ingest	Continuous	Devices publish every 30 s (spikes instantly); Lambda validates, de-duplicates, applies calibration	Clean readings in Timestream + PostGIS
Twin update	Every 15 min	GP interpolation + wind advection per 500 m cell	Cell values + confidence
Attribution	Every 15 min	Fingerprint rules + classifier; back-trajectory for top hotspots	Cause label + suspect pin

Stage	Trigger	Work done	Output
Forecast	Hourly (master 17:00)	LightGBM per horizon (1–72 h) with P10/P50/P90	Forecast grid + GRAP outlook
Plan	Every 3 h (master 18:30)	Exposure scoring → OR-Tools assignment → Planner and Auditor agents → explanations	Draft plan awaiting approval
Dispatch	On approval	Routes to drivers; alerts to schools and citizens; inspection tickets	Tasks in flight
Verify	2–4 h after each action	Synthetic control on treated cell vs donor cells	Effect size + confidence
Learn	Nightly	Retrain forecaster; update effectiveness library; recompute site scores	New model versions

8.2 Real-time path: a sudden spike

- **06:42** Bus 7 passes a vacant plot; its on-board TinyML flags PM2.5 jumping 3× in 20 s and publishes immediately.
- **06:43** Fingerprint engine: high PM2.5/PM10 ratio, VOC up, FIRMS shows no satellite fire (too small). Label: *likely open burning*.
- **06:44** A citizen photo from 300 m away is verified and matched. Confidence rises to high.
- **06:45** Planner proposes the nearest inspection team; Auditor confirms a sprinkler would not help smoke. Officer taps approve.
- **07:10** Inspector closes the case; the verifier tracks the cell returning to baseline and logs the outcome.

LAYERS, DATA MODEL, APIS, ALGORITHMS

9 Technical architecture



Highlighted = what no other team will have: intervention sensing, causal verification, exposure-first multi-agent planning

9.1 Layer by layer

Layer	Responsibility	Key design decisions
Edge	Sense, buffer, pre-filter	Same firmware for all nodes with role flags; X.509 per device; SD buffer for 4G gaps; spike events bypass batching
Ingest	Authenticate, validate, route	IoT Rules split telemetry vs events; malformed packets to a dead-letter queue; idempotent writes keyed by device + sequence
Store	Hot, warm, cold data	Timestream for raw series; PostGIS for cells, assets, POIs and plans; S3 Parquet for training; Redis for live twin tiles
Think	Turn readings into state	Python services in Lambda containers or SageMaker endpoints; every output carries uncertainty
Decide	Turn state into a plan	Deterministic optimiser proposes; LLM agents explain and critique but cannot bypass constraints
Act	Deliver to humans	Human-in-the-loop approval; WhatsApp/SMS for reach; open API for transparency

9.2 Core data model (PostgreSQL + PostGIS via Prisma)

Table	Key fields
device	id, role (M/S/TAG/EYE), vehicle_id, firmware, trust_score, cal_slope, cal_offset, cal_rh_coef, last_seen
reading (Timestream)	device_id, ts, geom, pm1, pm25, pm10, no2, co, voc, rh, temp, speed, seq
cell	id, geom (500 m polygon), ward, population, vulnerability_weight, land_use
cell_state	cell_id, ts, pm25_est, pm10_est, confidence, cause_label, cause_conf, exposure_ph

Table	Key fields
forecast	cell_id, issued_at, horizon_h, p10, p50, p90, grap_stage
asset	id, type (sprinkler/ASG/sweeper/team), depot_geom, capacity, shift, status
plan / plan_task	plan_id, status (draft/approved), created_by_agent, approved_by; task: asset_id, cell_id, eta, reason_text
intervention	id, task_id, asset_id, start_ts, end_ts, geom_track, water_litres, pm_before, pm_after
impact	intervention_id, effect_pct, ci_low, ci_high, duration_min, method, donor_cells
site	id, type, geom, grade (A/B/C), violations, last_inspection
report (citizen)	id, photo_s3, geom, verified, matched_spike, ticket_id, status

9.3 Main APIs (Node.js + TypeScript)

Endpoint	Purpose
GET /v1/twin?bbox&ts	Cell values and confidence as vector tiles for the map
GET /v1/forecast/:cellId	72 h P10/P50/P90 series and GRAP outlook
GET /v1/hotspots?ward	Ranked hotspots with cause, suspect pin and exposure
POST /v1/plans/draft	Run optimiser + agents for a ward and fleet; returns plan with reasons
POST /v1/plans/:id/simulate	What-If: compare this plan against a baseline in person-hours avoided
POST /v1/plans/:id/approve	Officer approval; dispatches tasks and alerts
GET /v1/interventions/:id/impact	Verified effect with confidence interval
POST /v1/reports (webhook)	WhatsApp Citizen Eyes intake
GET /open/v1/air	Public, rate-limited open data feed

9.4 Algorithms, in plain terms

Drive-by calibration (F3)

When a device is within 200 m of a reference station, moving slowly, with stable wind, pairs of readings are collected. A robust regression fits `ref = a·raw + b·RH + c` per device, which also corrects the known over-reading of optical PM sensors in humid air. Coefficients are versioned; a device whose fit keeps drifting loses trust score.

Digital twin (F4)

Calibrated points from the last hour are combined with a Gaussian-process (kriging) model using land-use covariates (road density, construction permits, industrial zones). The field is then shifted downwind by the IMD wind vector for the elapsed time. Each cell stores the posterior mean and variance; variance is the confidence shown to officers.

Source fingerprinting (F7)

Signal pattern	Likely cause	Right action
High PM10, low PM2.5/PM10 ratio, daytime, low NO_2, near roads or sites	Road / construction dust	Sprinkler, sweeper, site check
High NO_2 + CO, rush hours, along corridors	Traffic	Traffic diversion advice, smoky-vehicle checks, alerts
Sharp PM2.5 + VOC spike, night, fire point or photo	Open burning	Inspection team
City-wide PM2.5 rise, north-west wind, FIRMS fires upwind	Regional stubble smoke	No trucks; protect people, alerts

Exposure score (F12)

$E(cell, t) = \max(0, C - C_{safe}) \times Pop(cell, t) \times V(cell) \times \Delta t$, where *Pop* changes by hour (schools weighted in the morning, markets in the evening) and *V* is 1.0 for residential and higher near

schools, hospitals and elder-care homes. The planner minimises total forecast E.

Dispatch optimiser (F13)

A capacitated vehicle-routing problem in OR-Tools: each asset has a depot, shift window and capacity; each hotspot has a time window (before the forecast peak), a required asset type (from the cause) and a value equal to its expected exposure reduction, taken from the effectiveness library. The solver maximises exposure avoided within the shift; a greedy fallback runs in milliseconds.

Causal verification (F18)

For each intervention, donor cells are chosen that were not treated, have similar land use, and moved similarly over the previous six hours. A weighted synthetic control predicts what the treated cell would have looked like without the spray; the difference is the effect. A placebo test on donor cells gives a confidence interval, so a result like "PM10 −18% for 90 min" comes with its uncertainty.

CONSTRUCTION SHEET FOR ALL NODE TYPES

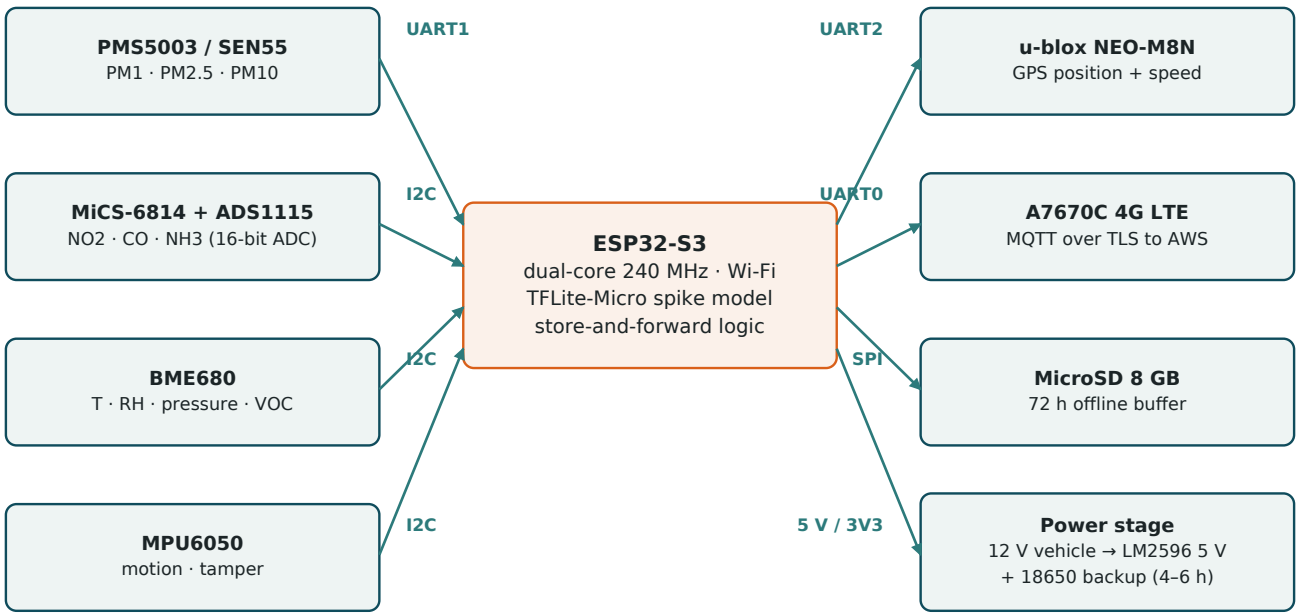
10 Hardware architecture

Four node types share one firmware codebase. For the hackathon we submit this specification and run a software simulator that emits the exact same packets; a physical HawaPod-M can be plugged in later with no cloud change.

Node	Where it lives	Job	Approx. unit cost
HawaPod-M (mobile)	Buses, e-rickshaws, delivery bikes	Street-level PM, gases, weather, GPS	₹7,000–9,500
HawaPod-S (sentinel)	School gate + classroom (a pair)	Outdoor vs indoor air, CO_2 for ventilation	₹4,500 (+ ₹2,500 CO_2)
HawaTag (intervention)	Sprinklers, smog guns, sweepers	Proves spraying happened; before/after PM	≈ ₹6,000
HawaEye (optional)	Selected buses at corridors	On-device smoky-vehicle detection	≈ ₹6,500

10.1 HawaPod-M block diagram

HawaPod-M block diagram · heated inlet + 5 V fan draw air from a forward-facing, roof-height tube



10.2 HawaPod-M bill of materials

#	Component	Purpose	Interface	Approx. ₹
1	ESP32-S3 DevKit (N16R8)	Controller, Wi-Fi, TinyML	—	600–800
2	Plantower PMS5003 (or Sensirion SEN55)	PM1 / PM2.5 / PM10 (laser scattering)	UART, 5 V	1,500–2,000
3	MiCS-6814 tri-gas sensor	NO_2, CO, NH_3	Analog → ADS1115	1,000–1,500
4	ADS1115 16-bit ADC	Clean analog reads (ESP32 ADC is noisy)	I2C	250
5	Bosch BME680	Temperature, humidity, pressure, VOC	I2C	900–1,200
6	u-blox NEO-M8N GPS + antenna	Position, speed, time	UART	350–700
7	SIMCom A7670C LTE Cat-1	Cellular MQTT uplink	UART	1,200–1,800
8	MPU6050 IMU	Motion state, tamper detection	I2C	150
9	MicroSD module + 8 GB card	Store-and-forward buffer	SPI	250
10	LM2596 buck (12 V → 5 V, 3 A)	Vehicle power	—	150
11	18650 cell + TP4056 + boost	Backup when ignition off	—	350
12	IP54 box, inlet tube, 5 V fan, heater coil	Weatherproof, controlled airflow	PWM (MOSFET)	400
	Total per unit			**≈ 7,000–9,500**

10.3 Wiring map (ESP32-S3, GPIO matrix allows remapping)

Module	Bus	Pins	Notes
PMS5003	UART1	TX 17 → RX, RX 18 ← TX, SET 8	Needs 5 V supply; 3.3 V logic is fine
GPS NEO-M8N	UART2	RX 16 ← TX, TX 15 → RX	1 PPS optional on GPIO 7 for time sync
A7670C 4G	UART0	TX 43, RX 44, PWRKEY 9	Separate 2 A rail; 1,000 µF bulk capacitor for TX bursts
BME680, MPU6050, ADS1115	I2C	SDA 4, SCL 5	Addresses 0x77, 0x68, 0x48
MiCS-6814 → ADS1115	Analog	A0 NO_2, A1 CO, A2 NH_3	Heater needs ~30 min warm-up; flag readings until stable
MicroSD	SPI	MOSI 11, MISO 13, SCK 12, CS 10	FAT32, ring buffer of 72 h
Inlet fan / heater	PWM	GPIO 6 / GPIO 21 via logic MOSFETs	Heater on when RH > 75%

10.4 Other nodes

HawaPod-S (school sentinel pair)

Same core board without 4G (uses school Wi-Fi) and mains powered. The indoor unit adds a **Senseair S8 NDIR CO_2** sensor (UART) so the system can say when to keep windows shut for particles versus when to ventilate for CO_2. The indoor/outdoor ratio proves whether an alert actually protected children.

HawaTag (intervention tracker)

Component	Purpose
ESP32-S3 + A7670C 4G + NEO-M8N GPS	Truck position and uplink
Hall-effect water flow sensor (sized for the pump line, e.g. 1" industrial)	Litres sprayed per minute: proves spraying, not just driving
ACS712 / non-invasive CT current sensor on pump motor	Confirms smog-gun fan and pump are running
PMS5003 on the cab roof	PM before and after each spray segment

Component	Purpose
Tamper seal switch + IMU	Detects removal or covering

HawaEye (optional smoky-vehicle node)

Raspberry Pi Zero 2 W with Camera Module 3 and a small YOLO model (int8) runs detection on the device and uploads only short evidence clips for flagged events, never continuous video. Clips go to an authorised review queue; nothing is published.

10.5 Firmware behaviour

- **Sampling:** PM every 5 s, gases every 10 s, GPS every 1 s; averaged and batched every 30 s.
- **Adaptive rate:** faster sampling inside key cells, slower in clean areas and when stationary (IMU), saving data and power.
- **TinyML spike detector:** a tiny model flags sudden spikes and publishes an event immediately on a priority topic.
- **Store-and-forward:** packets queued on SD when offline and replayed in order with sequence numbers.
- **Self-checks:** flatline, fan stall, heater fault and implausible jumps set health flags that feed the trust score.
- **OTA updates:** through AWS IoT Jobs, signed images, staged rollout with automatic rollback.

10.6 Packet format (MQTT topic hawa/v1/{deviceId}/telemetry)

```
{ "v":1, "dev":"HPM-0142", "role":"M", "seq":88214, "ts":"2026-11-12T01:14:30Z",
  "gps":{"lat":28.6331,"lon":77.2194,"spd":14.2,"hdop":0.9},
  "pm":{"pm1":141,"pm25":212,"pm10":318}, "gas":{"no2_ppb":61,"co_ppb":1840},
  "env":{"t":17.8,"rh":82,"p":1015,"voc_kohm":38}, "flags":["heater_on"],
  "fw":"1.4.2", "batt":4.02 }
```

10.7 Build rules (mistakes others make)

- Mount the inlet **forward-facing, above roof height**, and away from the vehicle's own exhaust, or you measure your own bus.
- Correct for **humidity**: optical PM sensors over-read in Delhi's winter fog. Heater in hardware plus RH term in calibration.
- Tag readings with **speed**; very high speed distorts inlet airflow, so the twin down-weights them.
- **Shade and ventilate** the enclosure; heat drift ruins electrochemical and metal-oxide gas readings.
- Treat low-cost gas sensors as **indicative**: use them for fingerprinting and ranking, not for legal compliance values.

EVERY TOOL, AND WHY IT IS THERE

11 Tech stack

Layer	Tools	Why
Firmware	ESP-IDF (C), TensorFlow Lite Micro, MQTT over TLS	Low power, on-device spike detection, secure uplink
Hardware sim	Node.js simulator on real OSM bus routes, AWS IoT Device SDK	Exercises the real pipeline with realistic noise, drift and humidity bias
IoT	AWS IoT Core, IoT Rules, IoT Jobs	Managed device identity, routing, OTA
Compute	AWS Lambda (Node.js + TypeScript), Lambda containers (Python)	Serverless, fits the free tier, scales with fleet
Streaming	Amazon SQS (hackathon) → Kinesis (scale)	Buffer bursts and replay
Storage	Amazon Timestream, RDS PostgreSQL + PostGIS, S3 (Parquet), DynamoDB	Time series, spatial queries, cheap history, device registry

Layer	Tools	Why
Cache / jobs	ElastiCache Redis + BullMQ	Live map tiles, planning job queue with retries
ORM / API	Prisma, Express or Fastify, Zod validation	Typed schema and contracts
ML	Python, scikit-learn (GP, DBSCAN), LightGBM, PyKrige, SageMaker	Twin, clustering, 72 h forecasting
Causal	Custom synthetic control (NumPy/SciPy) or pycausalimpact	Intervention effect with uncertainty
Optimisation	Google OR-Tools (CVRP with time windows)	Fleet routing and assignment
Agents	**Strands Agents SDK** (AWS open source) + Amazon Bedrock	Forecaster, Planner, Auditor, Communicator; satisfies the AWS open-source rule
Vision	Bedrock multimodal model; YOLO (int8) on HawaEye	Citizen photo checks, smoky-vehicle detection
Orchestration	Amazon EventBridge, AWS Step Functions	Scheduled cycles and multi-step workflows
Frontend	React + TypeScript, deck.gl + MapLibre, AWS Amplify Hosting	Fast heatmaps, trajectories, fleet animation
Auth	Amazon Cognito	Officer, driver and school roles
Messaging	Amazon SNS (SMS), WhatsApp Cloud API	Reach people where they already are
Ops	Amazon CloudWatch, AWS X-Ray, IaC with AWS CDK or SAM	Monitoring, tracing, reproducible deploys
External data	CPCB/DPCC, OpenAQ, IMD / ERA5, NASA FIRMS, WorldPop, OpenStreetMap	Ground truth, weather, fires, population, POIs

CAN THIS ACTUALLY BE BUILT AND DEPLOYED?

12 Feasibility

12.1 Technical

Every building block is proven: taxi and bus sensor networks have run in Shanghai, Guangzhou and Jinan; kriging and gradient-boosted forecasting are standard; OR-Tools routing is production-grade; synthetic control is a well-known causal method. The novelty is the **combination into one closed loop** and the **sensing of interventions**, not any unproven science.

12.2 Data availability

Data	Source	Status
Reference air quality	CPCB / DPCC stations, OpenAQ	Public, near real-time
Weather	IMD, ERA5 reanalysis	Public
Fire points	NASA FIRMS (VIIRS / MODIS)	Public, several passes a day
Population	WorldPop 100 m grids	Public
Schools, hospitals, roads	OpenStreetMap	Public
Bus routes	Delhi open transit data (GTFS)	Public
Fleet positions	Existing GPS on Delhi's machines	Requires agency integration in a pilot

12.3 Cost: pilot vs one smog tower

Item	Quantity	Estimated cost
HawaPod-M on buses/e-rickshaws	100	≈ ₹9.5 lakh
HawaPod-S pairs at schools	20 schools (40 units + 20 CO_2)	≈ ₹2.3 lakh

Item	Quantity	Estimated cost
HawaTag on sprinklers / smog guns	50	≈ ₹3 lakh
4G data (≈ 150 SIMs)	per year	≈ ₹3 lakh
AWS cloud (pilot scale)	per year	≈ ₹4–6 lakh (estimate)
Pilot total, year one		**≈ ₹22–24 lakh**
For comparison: one Delhi smog tower	build only	≈ ₹20–22 crore, plus about ₹15 lakh per month to run

Hardware prices are indicative Indian retail prices in 2026 and will change with volume and supplier; cloud cost is a planning estimate.

12.4 Regulatory and operational

- No airspace, chemical or medical approvals needed: sensors ride on vehicles, nothing is sprayed that is not already sprayed.
- Bus mounting needs transport-agency permission; a pilot can start with e-rickshaw fleets or a private delivery partner.
- Low-cost sensors are indicative, not regulatory-grade, so HawaCommand **targets and verifies**; official stations remain the legal record.
- HawaEye clips are processed on the device; only flagged clips are uploaded, retained briefly and reviewed by authorised staff.

12.5 What is real in the hackathon demo

Component	Status in demo
CPCB / OpenAQ, weather, FIRMS feeds	Live
IoT Core pipeline, storage, APIs, map	Live on AWS
HawaPod / HawaTag fleet	Simulated devices publishing real packets over real MQTT (clearly labelled)
Calibration, twin, fingerprint, forecast, exposure	Running on real historical and live reference data
Optimiser, What-If, agents, approval flow	Running
Causal verification	Running on simulated interventions with injected, known effects (so we can show it recovers them)
WhatsApp / SMS alerts	Live to test numbers

WHAT CHANGES FOR PEOPLE AND FOR THE CITY

13 Impact

Dimension	How HawaCommand creates it
Health	Children and outdoor workers warned before spikes with a concrete safe window; indoor/outdoor data shows the protection worked.
Environment	Machines go to cells where their cause-matched action works, at the time it matters, instead of fixed routes.
Money	Every rupee of dust control gets a measured effect. Ineffective deployments can be cut or moved, answering the evidence gap CEEW highlights.
Governance	Graded response starts 24–48 h early; site grading rewards good actors; anti-gaming checks keep numbers honest; open data builds trust, as AEROS did in Japan.
Equity	Mobile sensing covers areas with no station, where China's research shows unmonitored, often poorer residents were left out.

13.1 KPIs we will report (targets for a pilot, not claims)

KPI	Definition	Pilot target
Coverage	Share of pilot road-km observed at least daily	≥ 70%
Twin accuracy	Error of twin vs held-out reference station	Beat nearest-station baseline
Forecast skill	24 h PM2.5 error vs persistence forecast	Clear improvement over persistence
Exposure avoided	Person-hours above threshold avoided, AI plan vs usual routes (simulated, then measured)	Measured and reported weekly
Verified actions	Share of interventions with a verified effect and confidence interval	≥ 90%
Alert lead time	Hours between school alert and spike	≥ 12 h
Early action	GRAP-stage actions started before formal invocation	Tracked per episode

WHO PAYS, AND WHY THEY WOULD

14 Business model

Customer	Problem we solve for them	Offer	Indicative pricing (proposal)
Municipal bodies / pollution boards	Deploy fleets with evidence; show results to courts and citizens	Command centre SaaS per ward, HawaTag kits, evidence reports	Per-ward annual licence + hardware at cost
Large construction firms	Avoid stop-work orders; prove compliance	Site dust monitoring + compliance score + alerts	Monthly per site
Schools, school chains, campuses	Protect students; answer parents	HawaPod-S pairs + safe-window service	Annual per campus
RWAs and gated communities	Know local air; pressure the right agency	Neighbourhood sensors + dashboard	Annual subscription
Fleet operators (e-rickshaw, delivery)	Extra revenue, ESG reporting	Host sensors; share of data revenue	Revenue share
Researchers, insurers, health-tech	Street-level exposure data	Licensed historical data API	Usage-based

14.1 Go-to-market

- **Phase 1 (0–3 months):** one campus or district with an e-rickshaw fleet partner and 10–20 schools; publish verified results.
- **Phase 2 (3–12 months):** one MCD zone with HawaTags on its sprinklers and smog guns; integrate existing machine GPS.
- **Phase 3 (12+ months):** NCR-wide, then other NCAP cities; the effectiveness library becomes the moat.

14.2 Competition and positioning

Delhi already has forecasting and early-warning systems (for example IITM's), a Green War Room, GPS on machines, and private sensor networks and AQI apps. HawaCommand does not replace them. It is the **decision and verification layer** on top: it consumes their data, directs the fleet, and proves outcomes. No existing tool we found measures the effect of individual interventions.

WHAT COULD GO WRONG, AND THE ANSWER

15 Risks and mitigations

Risk	Mitigation
Low-cost sensors drift or read high in humidity	Drive-by calibration, heater, RH correction, trust scores, down-weighting of bad devices
Agencies will not adopt	Start with campuses, RWAs and private fleets that own their machines; publish verified results; integrate rather than replace
Source attribution is uncertain	Always shown with confidence; used to choose the tool, not to fine anyone; human approval on every action
Verification confounded by weather	Synthetic control with weather-matched donor cells, placebo tests, confidence intervals
LLM agents make a bad call	Agents only explain and critique; the optimiser enforces constraints; officer approval required
Privacy (cameras, locations)	On-device vision, evidence-only uploads, no faces or plates stored by default outside authorised review
Connectivity gaps	72 h store-and-forward, sequence numbers, idempotent ingest
Gaming (cleaning only near monitors)	Anti-gaming guard compares official stations with surrounding mobile readings
Hardware theft or vandalism	Tamper detection, IMU alerts, sealed enclosure, fleet-operator responsibility

24-HOUR SPRINT, TEAM OF FOUR

16 Hackathon build plan and demo

Hours	Backend lead	ML engineer	Frontend	Agents + hardware doc
0–3	AWS account, IoT Core, PostGIS schema, packet format	Pull CPCB, weather, FIRMS history	Map shell, deck.gl, Amplify	Finalise hardware sheet
3–8	Device simulator on OSM routes; ingest Lambda	Calibration + fingerprint rules	Twin heatmap layer	Strands agent skeleton
8–14	Exposure engine, plan APIs, BullMQ jobs	Digital twin + forecaster	What-If screen	Planner + Auditor with OR-Tools tool
14–19	Verifier service, EventBridge flows	Tune models; back-trajectory	Intervention view, approval flow	Communicator + WhatsApp/SNS alerts
19–22	Integration and bug bash (everyone)			
22–24	Demo video, README, AWS Builder Center blog draft (everyone)			

> **Cut rule.** If behind at hour 14, drop wind advection (keep plain kriging), drop Citizen Eyes and the school scheduler. **Never drop** the What-If screen and causal verification: they are the two things judges will not see anywhere else.

16.1 Three-minute demo video script

Time	Scene
0:00–0:20	The problem: one AQI number vs a street-level map of the same morning; 376 smog guns, and no one knows which ones work.
0:20–0:50	Live twin: buses streaming, hotspots appear with cause labels and upwind suspect pins.
0:50–1:20	72 h forecast shows a spike near three schools; GRAP outlook turns orange for Thursday.

Time	Scene
1:20–1:55	Agents draft a plan; the Auditor rejects a sprinkler sent to a smoke hotspot; What-If: AI plan vs usual routes.
1:55–2:20	Officer approves; driver route and school WhatsApp alert appear.
2:20–2:45	HawaTag logs a spray; the verifier shows the measured effect with a confidence interval.
2:45–3:00	Where AWS fits (architecture) and the closing line.

Closing line. "Delhi already owns the machines. HawaCommand gives them a brain, and gives the city proof. Same fleet, more people protected, and evidence for every rupee."

WHERE THE NUMBERS COME FROM

17 Sources

Claim	Source	Link
Delhi station count (47 CAAQMS, target 1 per 25 km²)	Millennium Post, May 2026	millenniumpost.in/delhi/delhi-to-get-13-new-caaqms-this-yr…
Delhi area 1,484 km² and station siting	Deccan Herald	deccanherald.com (Delhi government to relocate monitoring stations)
CAG audit on station placement	The Tribune	tribuneindia.com/news/delhi/cag-report-flags-defects-in-monitoring-air-quality-data
Fleet: 376 ASGs, 266 sprinklers, 91 sweepers, ~2,000 teams	The Tribune	tribuneindia.com/news/delhi/2k-teams-deployed-to-tackle-pollution-sirsa
Road dust ~15% of winter PM2.5; ~64% of NCAP funds	CEEW, Feb 2026	ceew.in/blogs/are-anti-smog-guns-the-solution-to-delhi's-air-pollution-problem
IIT Delhi finding on water spraying	Health Policy Watch, May 2026	healthpolicy-watch.news/delhi-has-a-new-plan-to-fight-its-toxic-air…
Smog towers: DPCC affidavit, 40,000 needed, ₹15 lakh/month	Down To Earth	downtoearth.org.in/pollution/smog-towers-ineffective…
Connaught Place tower cost and Xi'an link	Deccan Herald; The Asian Age	deccanherald.com; pressreader.com (The Asian Age, Sep 2021)
China hotspot grid, Cangzhou −38.3%	CAS Institute of Remote Sensing and Digital Earth	english.radi.cas.cn/News/UF/201809/t20180903_196899.html
Shanghai taxi network (125 taxis, 80–100 for 80% roads)	Sensors (MDPI) 2022 / PMC	pmc.ncbi.nlm.nih.gov/articles/PMC9416088
Guangzhou 314 taxis, roadside calibration	arXiv 2412.17831	arxiv.org/abs/2412.17831
Jinan 200 mobile + 614 micro-stations, 500 m/1 h maps	ResearchGate / ES&T 2026	researchgate.net/publication/407145342
IBM Green Horizons: 1 km, 72 h forecasts; Delhi what-if	IBM newsroom; MIT Technology Review	uk.newsroom.ibm.com (Dec 2015); technologyreview.com (Aug 2015)
Graded alerts ≥24 h ahead	Aerosol and Air Quality Research, 2020	aaqr.org/articles/aaqr-19-09-oa-0442
Early action 1–2 days before red alerts	Scientific Reports, 2017	nature.com/articles/s41598-017-08895-x
Performance grading and enforcement tools	Enviliance ASIA	enviliance.com/regions/east-asia/cn/cn-air/cn-heavy-pollution-weather
Smoky-vehicle detection; remote sensing in 70+ cities	Sensors 2023 (PMC); US patent 11639882	ncbi.nlm.nih.gov/pmc/articles/PMC10574957
Monitor-area improvement without city-wide drop	VoxDev	voxdev.org/topic/energy-environment/how-new-monitoring-systems-shaped-chinas-war-air-pollution
Japan AEROS / Soramame; per-pollutant alerts	HGPI; Smog Report	hgpi.org/en/research/ph-20251016.html; smogreport.com/guides/japan-aqi
Tokyo diesel rule: roadside SPM −20%+, compliance 0%→97%	Transportation Research Part D, 2008	sciencedirect.com/science/article/abs/pii/S136192090800028X
Environmental Hacks rules and judging	WeMakeDevs	wemakedevs.org/aws/env

Hardware prices, cloud costs and pilot KPIs are the author's estimates and targets, not measured results.